

Summary

- **Machine Learning Researcher** with expertise in developing **AI models**, having established a **pre-trained GAN** for **limited-labeled brain MRI segmentation**, advancing practical **AI** and **ML applications** and contributing to both **industry** and **research**.
- Passionate **AI Engineer** with over **5 years of experience** in **Computer Science**, possessing strong **technical**, **theoretical**, and **analytical** skills, including **model training**, **algorithm optimization**, and **problem-solving**.
- **Skilled Programmer** proficient in **Python** and other **programming languages**, along with modern **technologies**, with practical experience as an **iOS Developer** for an **international company**, demonstrating strong **technical expertise**, **teamwork**, and effective **communication** in **collaborative settings**.
- **Active Open-Source Contributor**, collaborating on **public** and **volunteer projects**, fostering **knowledge sharing** and providing **tools** for **developer** and **research communities**.

Education

Bachelor of Computer Engineering - GPA: 3.57/4.00 (equivalent to 17.88/20)
Zand Institute of Higher Education

Sep 2021 - Aug 2025
Shiraz, Iran

Visiting Student
University of Science and Culture

Feb 2023 - Jul 2024
Tehran, Iran

Research Interests

- Machine learning
- Computer Vision
- Deep Learning
- Computational Biology
- Natural Language Processing
- Scientific Computing

Technical Skills

- **Programming & Scripting:** Python, C/C++, Julia, MATLAB, Swift, Objective-C, SQL, Bash, LaTeX, ...
- **Deep Learning:** Neural Networks, Image Processing, Natural Language Processing, Transformers, GAN, RAG
- **Tools & Platforms:** Jupyter, Google Colab, GitHub, GitLab, Command Line, Linux, macOS, Jira, Confluence, Trello, VS Code, Xcode
- **Data Science & ML:** TensorFlow/Keras, PyTorch, Scikit-learn, OpenCV, HuggingFace, NumPy, Pandas, Matplotlib
- **Software Development:** Git, CI/CD, Design Patterns, Architectural Patterns, SOLID, iOS Development, Unit Testing, TDD, BDD, Agile (Scrum, Kanban)
- **Databases & Cloud:** MLOps, Vector DBs, Relational DBs (SQLite, PostgreSQL), Firebase, Docker, Kubernetes, Query optimization, Cloud-based AI Model Deployment

Publication

Pre-Trained Generative Adversarial Network for Limited-Labeled Brain MRI Segmentation

Authors: Mehdi Karami*, Dr. Betsabeh Tanoori
Code: To be released as open-source

*Corresponding author, Manuscript in preparation

Teaching Experience

Teaching Assistant (TA)

Zand Institute of Higher Education, Shiraz, Iran

Fall 2024:

- Human-Computer Interaction
- Digital Electronics

Spring 2025:

- Fundamentals of Computer and Programming
- Digital Electronics
- Computer Architecture

Work Experience

iOS Developer (Full-time, On-site)
Round Table Apps (Sydney, Australia)

Jun 2023 – Jun 2024
 Development Office, Tehran, Iran

- Engineered **Blossom – Save and Invest**, a consumer-facing financial application for the **Australian** and **New Zealand** markets, serving **25K+** active users.
- Maintained **MailPlus – MP Operator**, a post-transportation application for **MailPlus Australia**, supporting **300+** owner-operators and **30K+** customers.
- Developed and refactored **Resilience Box**, an **international** wellbeing and resilience platform.

Technical Contributions:

- Researched the development of **AR** and **VR** technologies in healthcare using **visionOS**.
- Conducted in-depth analysis of **data flows** to identify and model **user behavioral patterns** across various scenarios.
- Applied key design patterns such as **MVC**, **MVVM**, **VIP**, and **Repository**.
- Boosted development efficiency by up to **40%** through the creation of custom file templates.

Open Source Projects

- **Cyberattack Detection and Anomalous Behavior Analysis**
 - Utilizes an LSTM-based RNN to detect cyberattacks from anomalous network traffic, including DDoS, port scanning, and brute-force attempts.
 - Stack: TensorFlow/Keras, Scikit-learn, PCA, ROC Curve, ...
- **Solar Power Generation Prediction**
 - Implements ML models (Linear, Decision Tree, Random Forest, Gradient Boosting, MLP, DNN) to predict solar power generation, evaluated via MAE, MSE, R^2 , and confusion matrix.
 - Stack: TensorFlow, Scikit-learn, Pandas, NumPy, Seaborn, ...
- **NexumAI**
 - Provides an interface for training models (e.g., image classification, segmentation) without technical expertise, letting users obtain trained models by supplying datasets and configurations.
 - Stack: TensorFlow, PyTorch, Torchvision, PyQt6, ...
- **AI Sight Quest**
 - Utilizes AI to extract text from images via Apple's Vision Framework and provides instant answers to document questions using the Bidirectional Encoder Representations from Transformers (BERT) language model.
 - Stack: BERT-SQuAD, Swift, SwiftUI, SwiftData, Apple's Vision and Speech Frameworks, TipKit, Protocol-Oriented Programming (POP), Model-view-viewmodel (MVVM), ...
- **Numeric Prediction**
 - Utilizes collected car price records, applying preprocessing (e.g., normalization, standardization) to handle multilingual values, and implements KNN, Decision Tree, Random Forest, and Gradient Boosting algorithms.
- **Arduino Heart Rate Monitor (BPM Counter)**
 - Measures heart rate with an Arduino Uno and displays a real-time pulse waveform.
 - Stack: Arduino C/C++, Programmable Logic Controller (PLC), Sensor Integration, Display Handling, ...
- **Character Classification Using Perceptron**
 - Neural Network Design,
 - Forward/Backward Propagation,
 - Sigmoid, Softmax, Cross-Entropy, ...
 - Stack: pure NumPy
- **Information Retrieval**
 - Stopword Removal, Lemmatization,
 - Stemming, Tokenization,
 - Inverted Index, Boolean Model, ...
 - Stack: NLTK, inflect

Additional projects can be found at: [GitHub account](#).

Volunteering

Workshop on Git and GitHub

Zand Institute of Higher Education – [Certificate](#)

Dec 2024

K. N. Toosi University of Technology – [Certificate](#)

May 2024

Honors & Awards

Winning First Place in the C++ Programming Language Competition

May 2022

Zand Institute of Higher Education

Shiraz, Iran

Hobbies

- Swimming
- Traveling
- Cooking International Cuisine
- Esports Gaming (LOL)